

Department of Computer Science: Capstone Matrix & Syllabus

This curriculum document outlines the core course requirements, laboratory modules, and evaluation rubrics for the Senior Capstone Engineering Sequence (CS-490 / CS-491).

Course Code	Course Title	Lecture (Hrs)	Lab (Hrs)	Credits
CS-490	Senior Design Project I	3	3	4
CS-491	Senior Design Project II	1	6	4
CS-452	Cloud Computing & DevOps	3	2	4
CS-465	Distributed Algorithms	3	0	3
CS-480	Engineering Ethics & Law	2	0	2

Capstone Deliverables and Architectural Milestones

Students must execute projects in four-person agile teams. Deliverables include architectural design specifications, automated CI/CD pipelines, containerized deployment, and exhaustive integration test suites.

Milestone	Deliverable Description	Weight	Deadline
M1: Inception	Problem formulation, user stories, and system architecture document	15%	Week 4
M2: Prototype	Working MVP with containerized microservices and API documentation	25%	Week 8
M3: Verification	End-to-end integration tests, load testing, and security audit	25%	Week 12
M4: Final Defense	Live deployment, oral examination, and public code repository	35%	Week 16

Evaluation Rubrics and Performance Standards

Evaluation assesses code quality, architectural robustness, documentation rigor, and peer collaboration. Plagiarism or fabricated benchmarks results in immediate disqualification.

Performance Level	Technical Architecture	Code Quality & Tests	Team Collaboration
Exemplary (90-100%)	Fault-tolerant, scalable, clear microservices	95%+ branch test coverage, CI passing	Even commit distribution, active reviews
Proficient (75-89%)	Well-structured, minor coupling issues	80-84% test coverage, passing builds	Consistent participation across team
Developing (60-74%)	Monolithic tendencies, fragile state	60% test coverage, manual deployment	Imbalanced contributions, missed standups
Unacceptable (<60%)	Non-functional architecture, missing specs	No automated tests, broken code	Dysfunctional collaboration, missing code

Program Educational Outcomes & Accreditation Compliance

Upon graduation, students demonstrate capability to design complex engineering systems meeting realistic constraints including economic, environmental, ethical, health and safety, and sustainability considerations. All teams must archive project artifacts in institutional institutional repositories.